

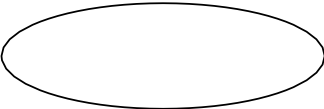
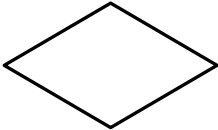


CS1002 – Programming Fundamentals

Lecture # 04
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FAST – NUCES, Faisalabad Campus

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Introduction to Flowcharts

Name	Symbol	Use in flowchart
Oval		Denotes the beginning or end of the program
Parallelogram		Denotes an input
Rectangle		Denotes a process to be carried out (e.g. addition, subtraction etc.)
Diamond/Decision		Denotes a decision (or branch) to be made. The program should continue along one of two routes. (e.g. IF/THEN/ELSE)
Flow Line		Denotes the direction of logic flow in the program
Off-page Connector		Denotes the connection of flow chart elements across pages

The Flowchart

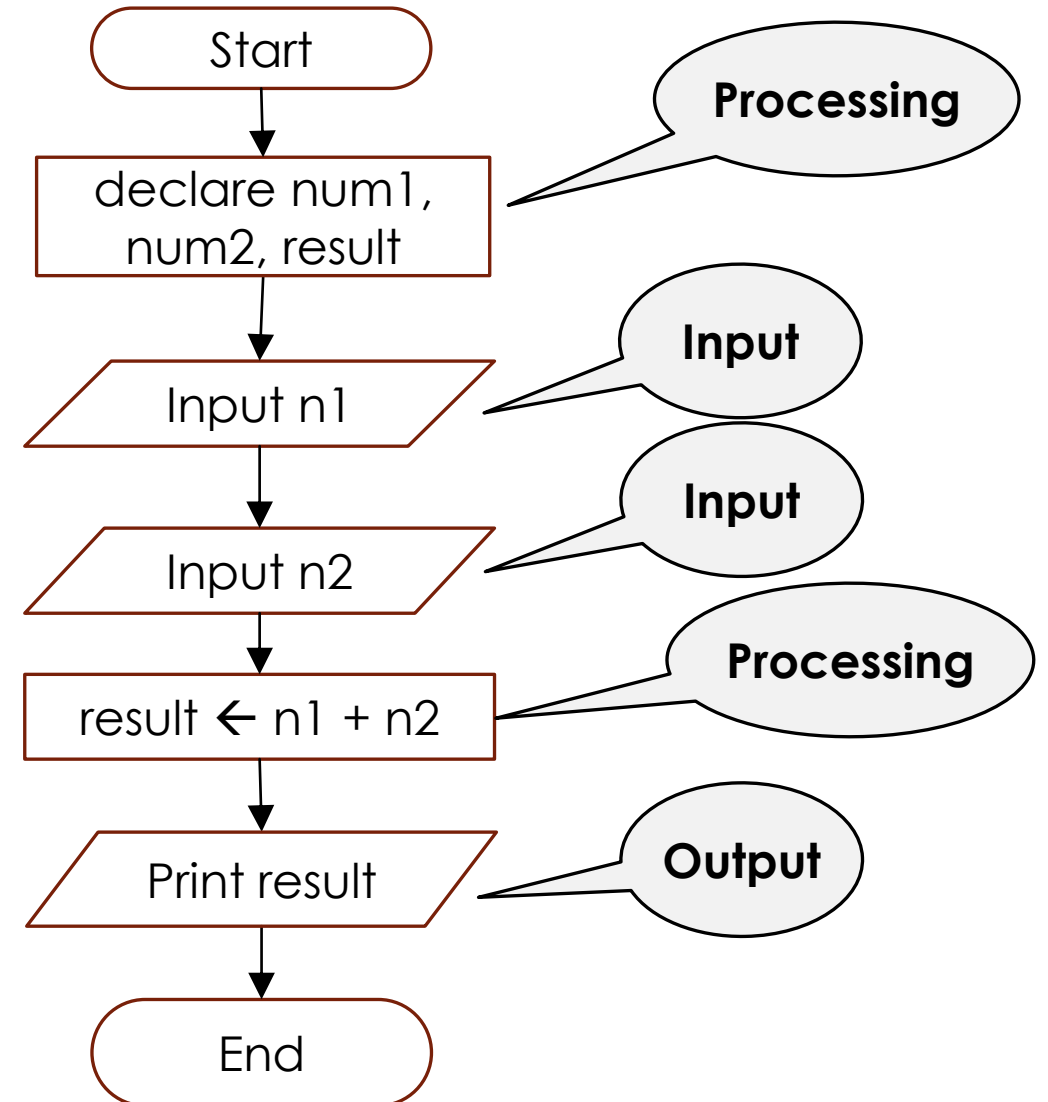
- **Dictionary Definition:** A schematic representation of a sequence of operation, as in a manufacturing process or computer program
- **Technical Definition:** A graphical representation of the sequence of operations in an information system or program
 - **Program flowcharts:** Shows the sequence of instructions in a single program or subroutine
- Different symbols are used to draw each type of flowchart

The Flowchart

- A flowchart
 - Shows logic of an algorithm
 - Emphasizes individual steps and their interconnections
 - E.g. control flow from one action to another

Example 1 (Pseudo code)

1. Start
2. declare num1, num2, result
3. input num1
4. input num2
5. results $\leftarrow$ num1 + num2
6. Print results
7. End

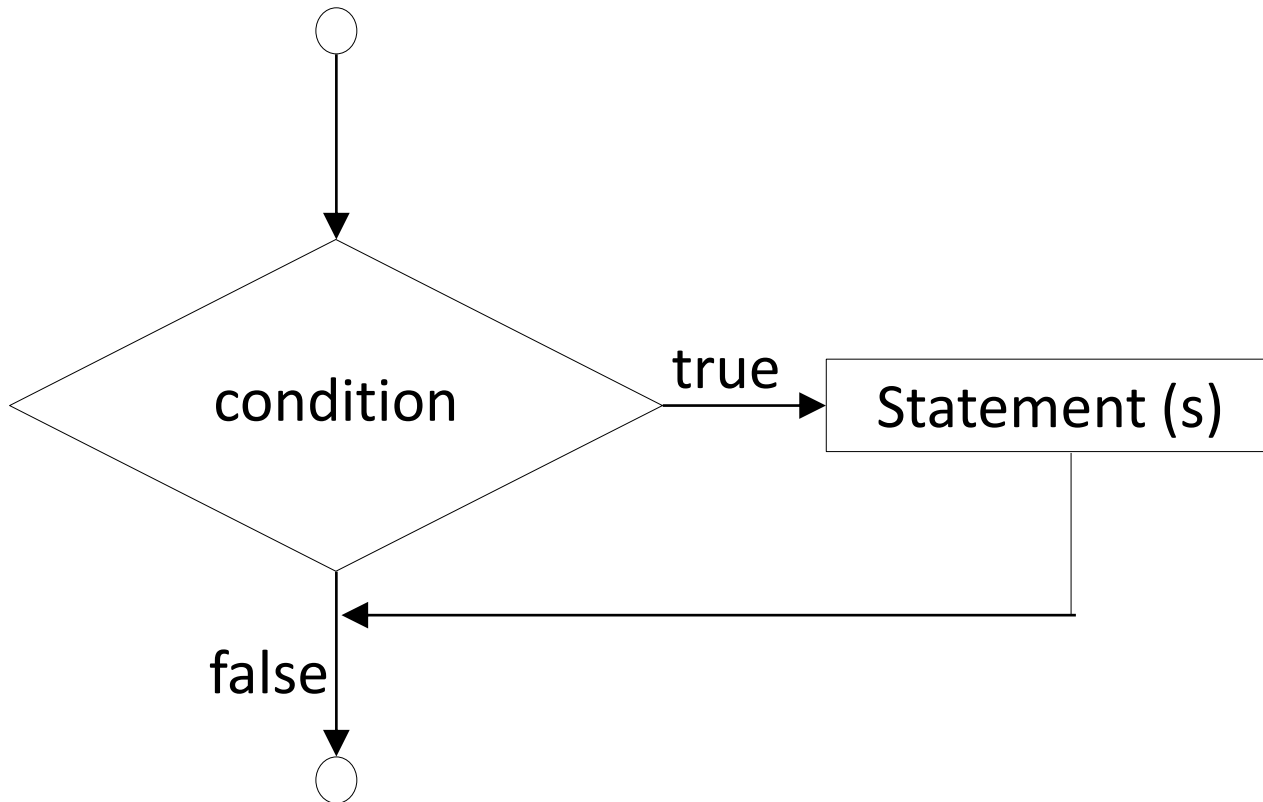


Class Task

- Draw flow chart for the following task
- Take 3 numbers from the user and print the average and sum of these three numbers

7 If selection structure

➤ Generic Format



A decision can be made on any expression.

zero - **false**

nonzero - **true**

Example:

3 - 4 is true

8 if Selection Structure

Translation into Algorithm

If student's grade is greater than or equal to 60
Print "Passed"

```
if ( marks >= 60 )  
    print "Passed";
```

➤ Diamond symbol (decision symbol)

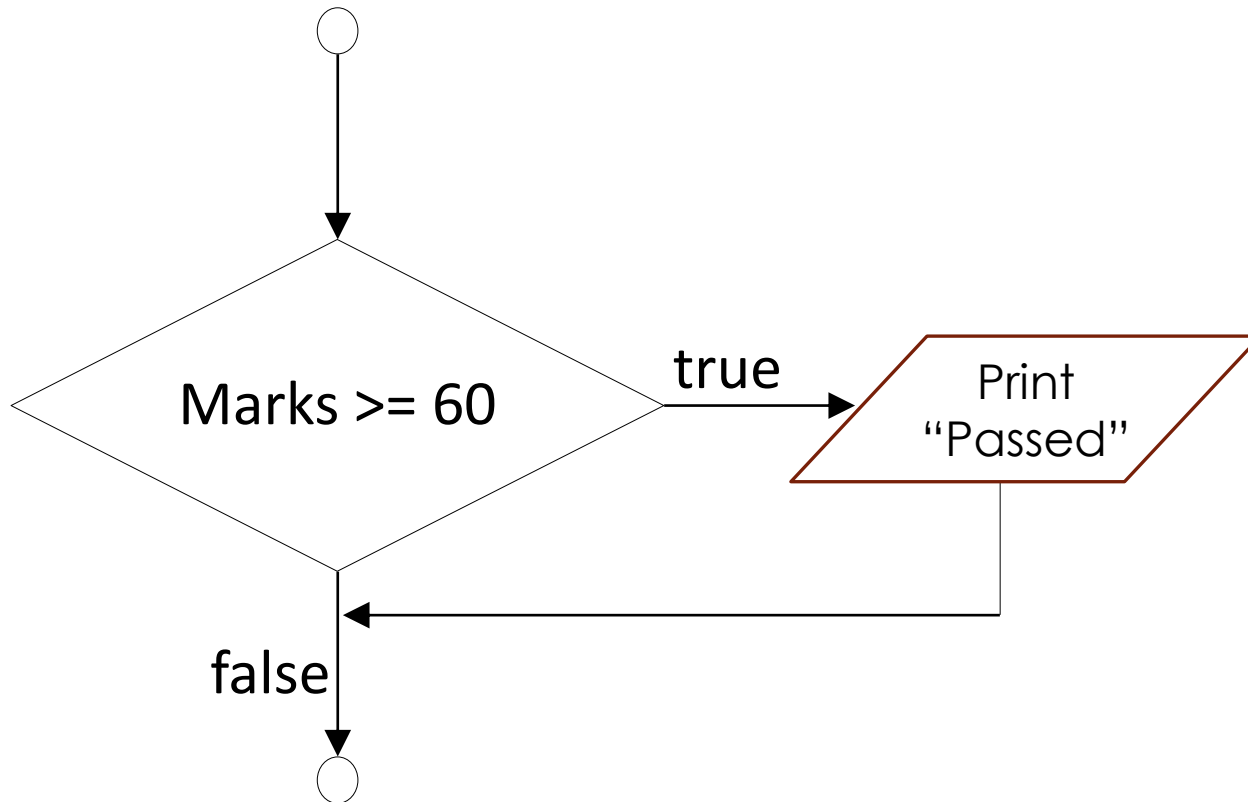
- Indicates decision is to be made
- Contains an expression that can be true or false
 - Test condition, follow path

➤ if structure

- Single-entry/single-exit

9 if Selection Structure

➔ Flowchart of pseudocode statement



A decision can be made on any expression.

zero - **false**

nonzero - **true**

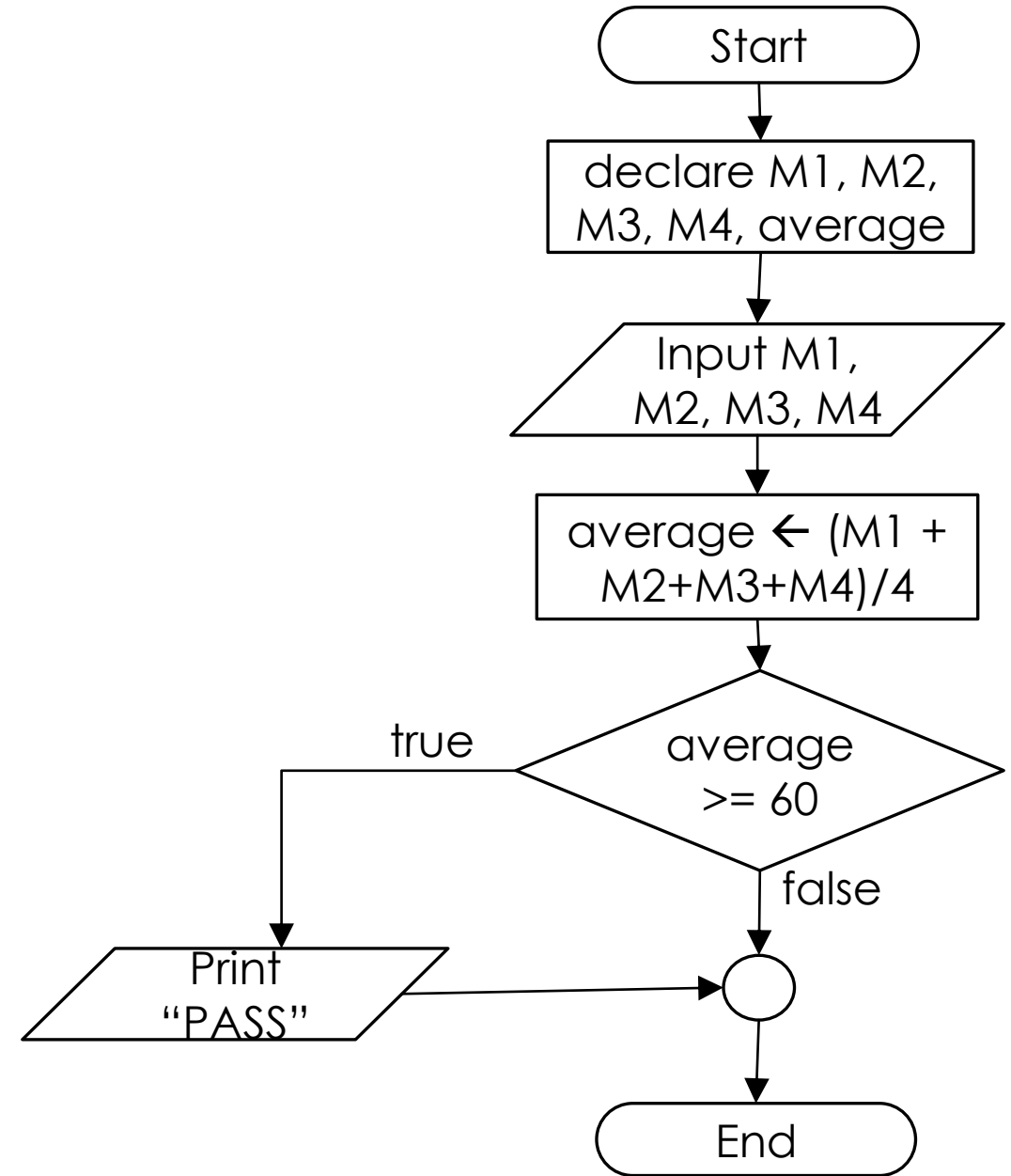
Example:

3 - 4 is true

Example 2: Flow chart

Pseudocode:

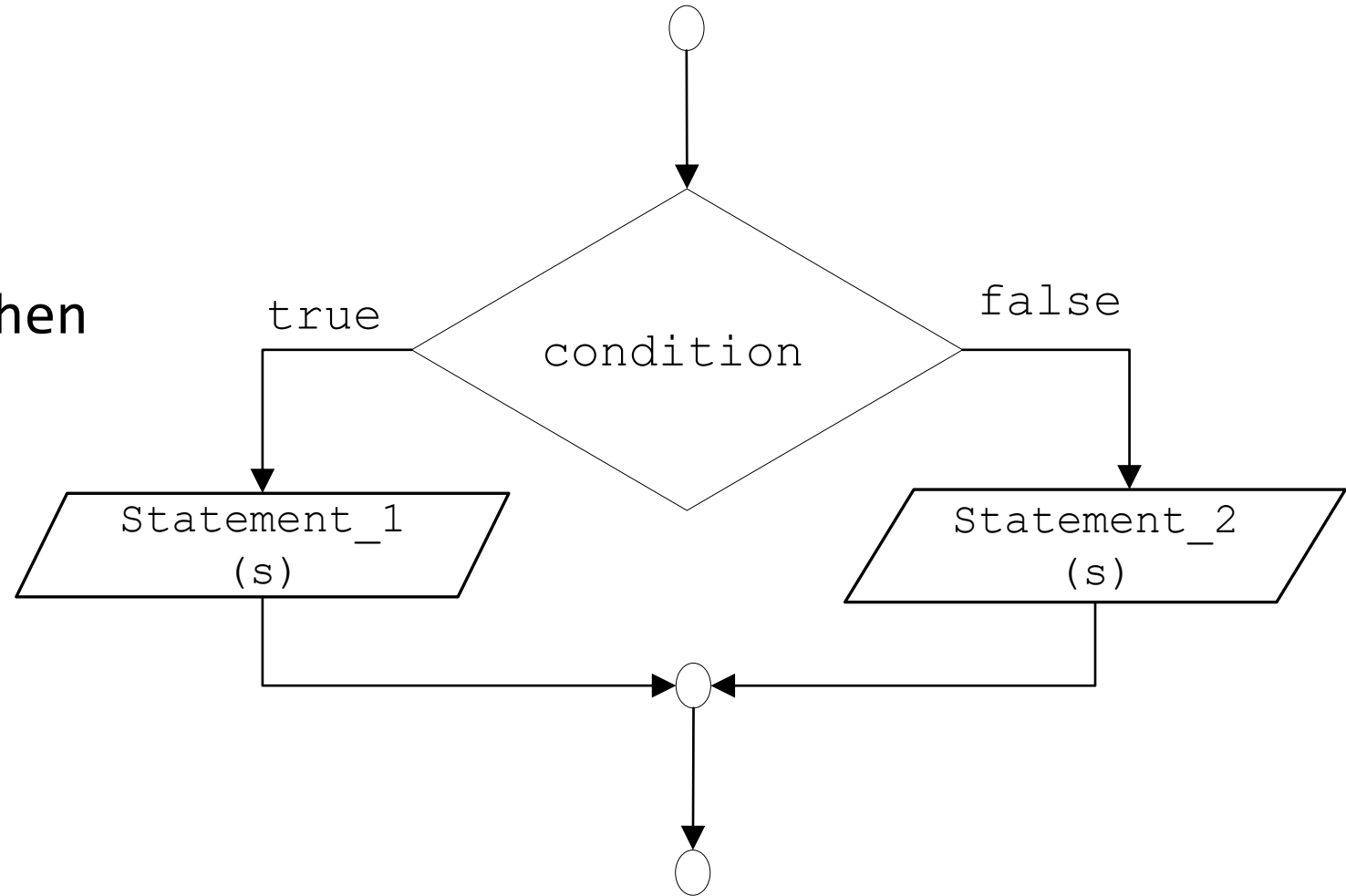
```
1.0 Start
2.0 Declare M1, M2, M3, M4, average
3.0 Input M1, M2, M3, M4
4.0 average = (M1+ M2+ M3+ M4) / 4
5.0 if (average > 50) then
    5.1 Print "PASS"
6.0 endif
7.0 End
```



if/else Selection Structure

➤ General Structure

```
if condition is met then  
    statement_1 (s)  
else  
    statement_2 (s)  
endif
```



if/else Selection Structure

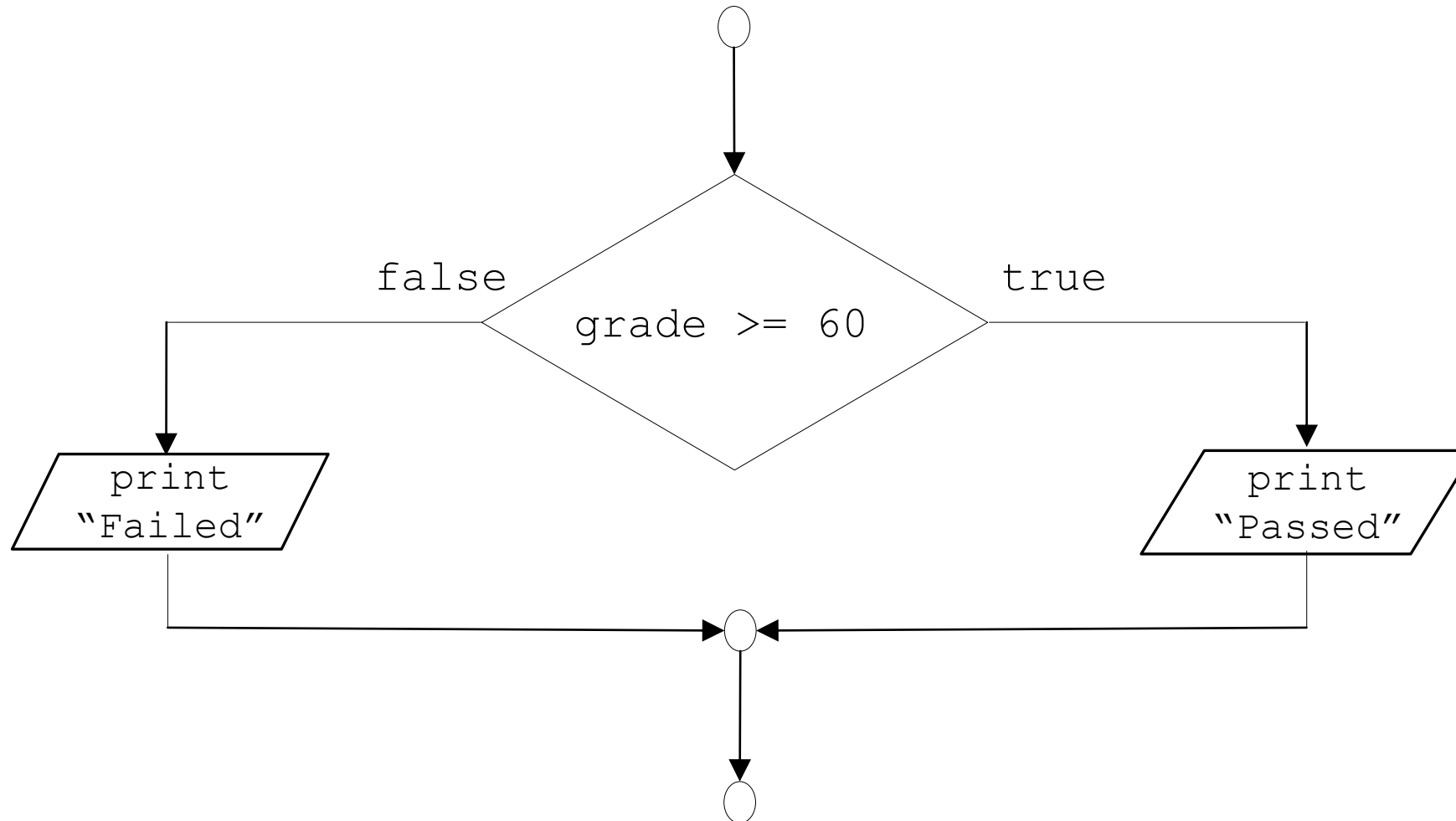
- if
 - Performs action if condition true
- if/else
 - Different actions if conditions true or false

Pseudocode

```
if student's grade is greater than or equal to 60
    print "Passed"
else
    print "Failed"
```

```
if ( grade >= 60 )
    Print "Passed";
else
    Print "Failed";
```

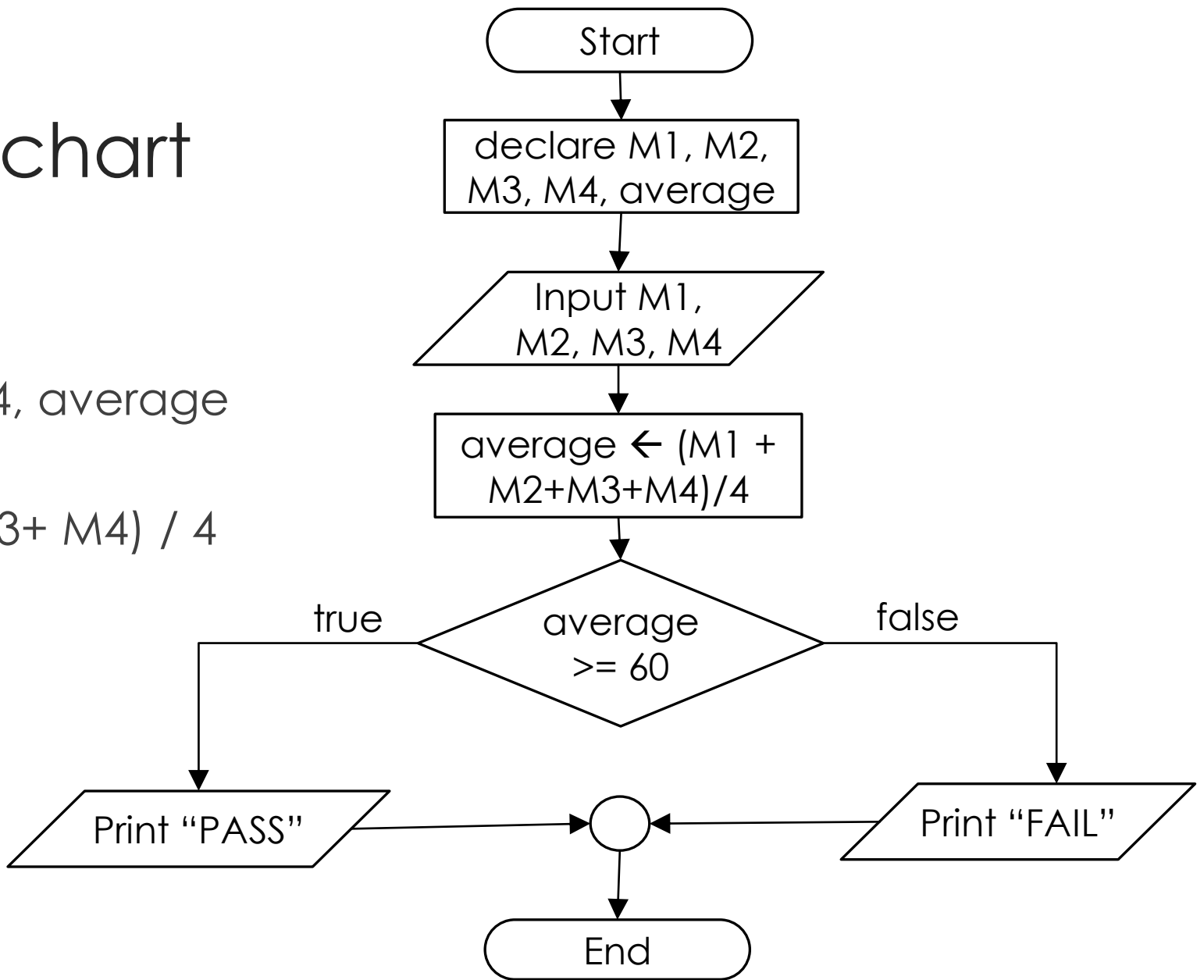
if/else Selection Structure



Example 3: Flow chart

Pseudocode:

```
1.0 Start
2.0 Declare M1, M2, M3, M4, average
3.0 Input M1, M2, M3, M4
4.0 average = (M1 + M2 + M3 + M4) / 4
5.0 if (average > 50) then
    5.1 Print "PASS"
6.0 else
    6.1 Print "FAIL"
7.0 endif
8.0 End
```



Example 4

- Write a pseudocode and draw a flowchart to convert the length in feet to centimeter
- Algorithm:
 - Input the length in feet (*Lft*)
 - Calculate the length in cm (*Lcm*) by multiplying *Lft* with 30
 - Print length in cm (*Lcm*)

Example 5

- Draw a flowchart to calculate area of a rectangle
- The program should ask the user to input Length and Width and then display the Area.
- $\text{Area} = \text{Length} * \text{Width}$

Example 6

Draw a flowchart that reads two values, determines the largest value and prints the largest value with an identifying message

Pseudocode

```
1.0 Declare VALUE1, VALUE2, MAX
2.0 Input VALUE1, VALUE2
3.0 if (VALUE1 > VALUE2) then
3.1     MAX ← VALUE1
4.0 else
4.1     MAX ← VALUE2
5.0 end if
6.0 Print "The largest value is", MAX
```

Questions

